Stochastic approach to de Sitter phase transition and interacting fields

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1. Introduction

The stochastic formalism provides a straightforward approach for computations within de Sitter space-time. It employs a stochastic Langevin equation to elucidate the behavior of the long-wavelength field modes, incorporating a noise component originating from the short-wavelength quantum modes and the rate at which the metastable vacuum decays is determined by the smallest non-zero eigenvalue linked to the Fokker-Planck equation. The spectral expansion method offers a potent approach to solving stochastic systems by utilizing the eigenvalues and eigenfunctions of a Schrödinger-like equation. This approach allows us to gain an understanding of the phase transition in de Sitter space-time from a different point of view.

2. Stochastic Formalism

The Langevin equation serves as the starting point for the stochastic approach, $\phi(X,t)$ is the scalar field and $f(X,t)$ is noise[1]:

$$\dot{\phi}(X,t) = -\frac{1}{3H}\frac{dV(\phi)}{d\phi} + f(X,t) \tag{1}$$

Leading us to the Fokker-Planck equation with $\rho(\phi,t)$ as the one point distribution function[2]:

$$\boxed{\frac{\partial\rho(\phi,t)}{\partial t} = \left(\frac{V''(\phi)}{3H} + \frac{V'(\phi)}{3H}\frac{\partial}{\partial\phi} + \frac{H^3}{8\pi^2}\frac{\partial^2}{\partial\phi^2}\right)\rho(\phi,t)} \tag{2}$$

Its solution in terms of the orthogonal set of eigenfunctions of the Schrödinger-type equation is:

$$\boxed{\rho(\phi,t) = e^{\left(-\frac{4\pi^2 V(\phi)}{3H^4}\right)}\sum a_n\Phi_n(\phi)e^{-\Lambda_n(t-t_0)}} \tag{3}$$

$$-\frac{1}{2}\frac{\partial^2\Phi_n(\phi)}{\partial\phi^2} + \frac{1}{2}\left[\left(\frac{4\pi^2}{3H^4}\right)^2 V'(\phi)^2 - \left(\frac{4\pi^2}{3H^4}\right)V''(\phi)\right]\Phi_n(\phi) = \frac{4\pi^2\Lambda_n}{H^3}\Phi_n(\phi) \tag{4}$$

3. Correlation Functions

The two-point spatial correlation function at equal time can be written as[3]:

$$\langle\phi(X_1,t)\phi(X_2,t)\rangle = \int d\phi_1 d\phi_2 \phi_1\phi_2\rho_2(\phi_1,\phi_2,t) \tag{5}$$

In which $\rho_2(\phi_1,\phi_2,t)$ is the two point probability distribution function.
Also the temporal correlation function is:

$$\langle\phi(X,t_1)\phi(X,t_2)\rangle = \int d\phi_1 d\phi_2 \phi_1\phi_2\rho_2(\phi_1,\phi_2,\tilde{t}) \tag{6}$$

Each correlation function can be found by calculating ρ_2 as the function satisfying two-point Fokker-Planck equation with $\tilde{t} = |t_1 - t_2|$.

4. Interacting Fields

Considering a potential with quadratic and quartic terms, under the $m^2 > 0$ condition , the Schrödinger-like equation expressed with only dimensionless parameter $\alpha = \frac{m^2}{\sqrt{\lambda}H^2}$ and dependency on new variable $\beta = \frac{\lambda^{\frac{1}{4}}}{H}\phi$ becomes($\Phi_n(\phi)$ turns into $\Phi_n(\alpha;\beta)$)[4]:

$$\boxed{\left\{\frac{\partial^2}{\partial\beta^2} - \left(\frac{4\pi^2}{3}\right)^2\beta^2(\alpha+\beta^2)^2 + \frac{4\pi^2}{3}(\alpha+3\beta^2)\right\}\Phi_n(\alpha;\beta) = -8\pi^2\frac{\Lambda_n(\alpha)}{\lambda^{\frac{1}{2}}H}\Phi_n(\alpha;\beta)} \tag{7}$$

We can find the eigenvalues as perturbative expansions in powers of α for the massless limit($\alpha << 1$) and the free limit ($\alpha >> 1$), resulting in:

Λ_n	$\alpha << 1$	$\alpha >> 1$
$n=0$	0	0
$n=1$	$0.088\sqrt{\lambda}H + 0.214\sqrt{\lambda}H\alpha + O(\alpha^2)$	$\frac{m^2}{H}(\frac{1}{3} + \frac{3}{8\pi^2\alpha^2}) + O(\alpha^{-4})$
$n=2$	$0.289\sqrt{\lambda}H + 0.351\sqrt{\lambda}H\alpha + O(\alpha^2)$	$\frac{m^2}{H}(\frac{2}{3} + \frac{3}{2\pi^2\alpha^2}) + O(\alpha^{-4})$
$n=3$	$0.536\sqrt{\lambda}H + 0.529\sqrt{\lambda}H\alpha + O(\alpha^2)$	$\frac{m^2}{H}(\frac{4}{3} + \frac{6}{\pi^2\alpha^2}) + O(\alpha^{-4})$

5. Vacuum Decays

•In **bounded potentials**: Introducing the concept of decreasing probability of remaining in the local minimum as $p_1(t) = e^{-\Gamma(t-t_0)}$, Then[5]:

field's PDF — global minimum's (true vacuum) PDF — local minimum's (false vacuum) PDF

$$\rho(\phi,t) = (1-p_1(t))\rho_{eq}(\phi) + p_1(t)\rho_1(\phi) \tag{8}$$

▷ The false vacuum decay rate is defined as the lowest non-zero eigenvalue:

$$\boxed{\Gamma = \Lambda_1} \tag{9}$$

• In **unbounded potentials**: $\Phi_0(\phi)$ does not meet the boundary condition, following eq.(3) $\Longrightarrow \rho(\phi,t) = e^{-\Lambda_1(t-t_0)}\frac{1}{N}\Phi_0(\phi)\Phi_1(\phi)$.
▷ Once more, the vacuum decay rate is determined by the lowest eigenvalue:

$$\boxed{\Gamma = \Lambda_1} \tag{10}$$

6. Conclusion

The stochastic formalism can describe how vacuum decay happens in de Sitter space in a way that the fields are seen as variables, with short wavelengths (small-scale fluctuations) adding Gaussian noise to the long wavelengths and the correlation functions can be found from the probability density function. The large-scale scalar fields can be described by the Fokker-Planck equation. In this formalism, the decay rate of the vacuum is determined by the lowest non-zero eigenvalue of the Schrödinger-like equation allowing us to develop a phenomenological explanation for the vacuum decay and phase transition within the inflationary state. In order to have a deeper understanding, it's essential that the calculations and the formalism should also be examined under the quantum field theory considerations.

7. References

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